



University of Tripoli

Faculty of Engineering

Department of Biomedical Engineering

Control and Instrumentation Division

Voice Signal Analysis Using Machine Learning for Early Detection of Parkinson's Disease

**A Project Report Submitted in Partial Fulfillment of the Requirements for the
Degree of Bachelor of Science in Biomedical Engineering**

Prepared by:

Suhail Mohamed Alhammali

ID: 2200208522

Supervised by:

Dr. Adel Adyaf

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Tripoli – Libya

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

﴿وَقُلْ رَبِّ زِدْنِي عِلْمًا﴾

سورة طه، الآية 114

الإهداء

إلى والديّ، اللذين كان دعاؤهما وصبرهما سنداً لي في كل خطوة.
وإلى أساتذتي في قسم الهندسة الطبية الحيوية، اللذين علّموني أن السؤال الصحيح أهمّ من الجواب السريع.
وإلى كل من يعمل على أن تصبح الرعاية الصحية أقرب متناولاً وأقلّ كلفة.

Declaration

I hereby declare that the project report entitled **Voice Signal Analysis Using Machine Learning for Early Detection of Parkinson's Disease**, submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Science in Biomedical Engineering from the Department of Biomedical Engineering at the University of Tripoli, is a bona fide work carried out by me under the supervision of **Dr. Adel Adyaf**.

This submission represents my ideas in my own words, and where the ideas or words of others have been included, I have adequately and accurately cited and referenced the original sources. I also declare that I have adhered to the ethics of academic honesty and integrity and have not misrepresented or fabricated any data, ideas, facts, or sources in my submission. I understand that any violation of the above may result in disciplinary action by the institute and/or the university, and may also give rise to legal action by rights holders whose work has not been properly cited or for which permission has not been obtained. This report has not previously formed the basis for the award of any degree, diploma, or similar title at any other university.

Suhail Mohamed Alhammali

Spring 2026

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Abstract

[Draft placeholder — the abstract will be finalized after all chapters are approved.]

Parkinson’s disease alters the muscles used in speech production, producing measurable changes in the voice. This project develops a complete, reproducible research prototype that analyses raw voice recordings for Parkinson’s-related acoustic patterns using classical, explainable machine learning. Seventy-four acoustic features covering pitch statistics, jitter, shimmer, harmonics-to-noise ratio, cepstral peak prominence, pause behaviour, and spectral shape are extracted from 10-second segments of each recording and averaged. A Random Forest classifier, validated with strictly subject-independent cross-validation on the MDVR-KCL dataset (37 subjects), achieves a subject-level balanced accuracy of 0.822 ± 0.032 . External validation on an independent Italian dataset with no adaptation yields a speaker-level area under the receiver operating characteristic curve of 0.701, quantifying generalization across language and recording conditions. A screening prototype with deliberately cautious, non-diagnostic reporting completes the system.

الملخص

يؤثر مرض باركنسون في العضلات المسؤولة عن إنتاج الكلام، مما يُحدث تغيّرات قابلة للقياس في الصوت. يقدّم هذا المشروع نموذجاً بحثياً أولياً متكاملًا وقابلًا لإعادة الإنتاج يحلّل التسجيلات الصوتية الخام بحثاً عن الأنماط الصوتية المرتبطة بمرض باركنسون باستخدام تعلّم الآلة التقليدي القابل للتفسير. تُستخلص أربع وسبعون خاصية صوتية تغطي إحصاءات التردد الأساسي والارتجاج والتذبذب ونسبة التوافقيات إلى الضجيج وبرزوز القمة الكبستراكية وسلوك التوقّفات وشكل الطيف، من مقاطع مدّتها عشر ثوانٍ من كل تسجيل ثم يُؤخذ متوسطها. حقّق مصنّف الغابة العشوائية، بعد التحقّق المتقاطع المستقل تماماً عن المتحدث على قاعدة البيانات MDVR-KCL (سبعة وثلاثون شخصاً)، دقّةً متوازنة على مستوى الشخص قدرها 822.0 بانحراف معياري 032.0. كما أسفر التحقّق الخارجي على قاعدة بيانات إيطالية مستقلة دون أي تكييف عن مساحة تحت منحنى خصائص التشغيل للمستقبل قدرها 701.0 على مستوى المتحدث، بما يقيس قابلية التعميم عبر اللغة وظروف التسجيل. ويكتمل النظام بنموذج فرز أولي يعتمد صياغة حذرة غير تشخيصية.

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List of Abbreviations

Abbreviation	Definition
AUC	Area Under the Curve
CLI	Command-Line Interface
CPPS	Smoothed Cepstral Peak Prominence
CSV	Comma-Separated Values
CV	Cross-Validation
DC	Direct Current
F0	Fundamental Frequency
FN	False Negative
FP	False Positive
HC	Healthy Control
HNR	Harmonics-to-Noise Ratio
MDS	Movement Disorder Society
MDVR-KCL	Mobile Device Voice Recordings at King's College London
MFCC	Mel-Frequency Cepstral Coefficient
ML	Machine Learning
MLP	Multilayer Perceptron
PD	Parkinson's Disease
RBF	Radial Basis Function
RF	Random Forest
RMS	Root Mean Square
ROC	Receiver Operating Characteristic
SNR	Signal-to-Noise Ratio
SVM	Support Vector Machine
TN	True Negative

TP	True Positive
WAV	Waveform Audio File Format

Chapter 1

Introduction

1.1 Overview

This report documents the design, implementation, and evaluation of a research prototype that analyses raw voice recordings for acoustic patterns associated with Parkinson’s Disease (PD) using classical, explainable Machine Learning (ML). The system covers the complete chain from an audio file to a cautious screening indication: signal preprocessing, acoustic feature extraction, model training, subject-independent validation, external validation on an independent dataset, and a file-based screening application. This chapter introduces the medical and engineering motivation behind the project, states the problem formally, lists the project objectives, defines the scope and the non-diagnostic framing of the system, and outlines the organization of the remainder of the report.

1.2 Background and Motivation

Parkinson’s Disease is the second most common neurodegenerative disorder after Alzheimer’s disease. It is caused by the progressive loss of dopamine-producing neurons in regions of the brain that control movement, and it manifests as tremor, muscular rigidity, slowness of movement, and reduced movement amplitude. The disease burden is large and growing: the Global Burden of Disease study estimated that about 6.1 million people were living with PD in 2016, more than double the number in 1990, a rise driven mainly by population ageing [1].

Despite this burden, no simple objective test for PD is available in routine practice.

The diagnosis remains clinical: it is established from motor signs observed by a neurologist, following criteria such as those of the Movement Disorder Society (MDS) [2]. Post-mortem morphometric evidence indicates that a substantial proportion of the dopaminergic neurons of the substantia nigra is typically already lost by the time the characteristic motor signs appear [3], which narrows the window in which future disease-modifying treatments could be most useful. These facts together motivate the search for inexpensive, non-invasive measurements that could support earlier referral of at-risk individuals to clinical evaluation [2, 3].

The human voice is a promising carrier of such measurements. Speaking is a motor activity that depends on the coordinated action of the respiratory muscles, the vocal folds of the larynx, and the articulators of the vocal tract; the same neuromuscular impairments that affect limb movement in PD also affect these structures. Speech and voice impairment — reduced loudness, monotone pitch, breathiness, and imprecise articulation, collectively described as hypokinetic dysarthria — has been reported in approximately 90% of a large clinical sample of PD patients [4]. Importantly for screening, acoustic abnormalities appear early: quantitative acoustic analysis detected some form of vocal impairment in about 78% of patients with early, untreated PD [5], and a longitudinal case study found measurable reduction of fundamental-frequency variability in recordings made about five years before the clinical diagnosis [6].

A voice recording can be captured in seconds with a commodity microphone or smartphone, at negligible cost, with no discomfort, and repeatedly. If acoustic measurements of such recordings can be related to PD status with useful reliability, voice analysis could serve as a low-cost screening-support signal, flagging individuals for whom a clinical examination is advisable. This prospect has produced a large research literature and several public datasets, including the Mobile Device Voice Recordings at King’s College London (MDVR-KCL) corpus used in this project [7].

Two considerations shaped the engineering approach taken here. First, in a biomedical setting, the ability to explain *why* a model produced its output is valued alongside raw performance; classical ML models operating on physiologically interpretable acoustic features were therefore preferred over deep learning, whose learned representations are difficult to interpret and whose data requirements far exceed the 37 subjects available. Second, published results in this field frequently report accuracies above 95%, which, as

examined in Chapters 2 and 4, often stems from evaluation protocols that let the same speaker appear in both training and test data. Producing an honest, subject-independent, and externally tested estimate of performance was therefore treated not as an obstacle but as a central contribution of the project.

1.3 Problem Statement

The problem addressed by this project is stated as follows: given a raw voice recording of continuous speech, determine — by means of reproducible signal processing and classical machine learning — whether the acoustic pattern of the recording is closer to that of a group of speakers with PD or to that of a group of healthy control speakers, and report the result with an honest, subject-independent estimate of reliability and with wording appropriate to a non-diagnostic research tool.

Solving this problem requires answering several subsidiary questions: which acoustic measurements meaningfully capture Parkinson’s-related voice changes in continuous speech; how such measurements must be extracted so that training, evaluation, and prediction remain strictly consistent; how a model must be validated when several recordings originate from the same person; how much of the measured performance survives transfer to a different dataset, language, and recording channel; and how the output of such a system must be presented to avoid overstating what it knows.

1.4 Project Objectives

The objectives were divided into a primary set, which defines the minimum complete system, and an extended set, which was undertaken after the primary objectives were met.

1.4.1 Primary Objectives

1. Inspect and verify the MDVR-KCL dataset before any modelling: file integrity, class labels, subject identities, and the feasibility of subject-independent validation.
2. Implement a consistent preprocessing pipeline for raw audio (mono conversion, fixed sample rate, offset removal, silence trimming, and amplitude normalization)

that deliberately avoids operations that would distort clinically meaningful signal irregularities.

3. Implement, as a core contribution, the extraction of physiologically motivated acoustic features: fundamental frequency statistics, jitter, shimmer, harmonics-to-noise ratio, and mel-frequency cepstral coefficient statistics.
4. Train and compare classical ML classifiers (logistic regression, support vector machine, random forest, and multilayer perceptron) against a majority-class baseline.
5. Evaluate all models with strictly subject-independent cross-validation, reporting balanced accuracy, sensitivity, specificity, F1 score, confusion matrices, and the area under the receiver operating characteristic curve, together with their variability across folds and random seeds.
6. Build a file-based screening prototype (browser application and command-line interface) that reuses the identical pipeline and presents results with mandatory non-diagnostic wording.

1.4.2 Extended Objectives

7. Improve classification performance through recording segmentation and additional continuous-speech features (smoothed cepstral peak prominence, pause statistics, and delta mel-frequency cepstral coefficients) under model-selection rules declared before experimentation, so that the improvement process itself cannot overfit the validation data.
8. Demonstrate quantitatively, on the project’s own data, how an incorrect validation design inflates reported performance.
9. Validate the final, frozen model externally on an independent dataset recorded in a different language and under different conditions, and report the resulting generalization gap.
10. Extend the prototype with uncertainty-aware reporting: an inconclusive score band, recording-condition warnings, and a microphone demonstration mode explicitly

framed as out-of-domain.

All ten objectives were achieved; the evidence is presented in Chapter 4.

1.5 Scope and Non-Diagnostic Framing

The system developed in this project is a research screening-support prototype. Throughout this report, and inside the software itself, its output is described as a non-diagnostic indication of similarity between the acoustic pattern of a recording and the patterns of the dataset groups. The system does not diagnose Parkinson’s Disease, does not measure any person’s medical risk, and must not replace evaluation by a qualified healthcare professional. Many conditions other than PD — among them common respiratory infections, ageing, smoking, and fatigue — also alter voice quality, and the datasets used are far too small and too homogeneous to support any clinical claim. The quantitative results presented in this report characterize performance over the datasets studied, not the accuracy of any statement about an individual. A dedicated discussion of limitations and the ethical constraints applied to the system’s wording is given in Section 4.12.

Within these boundaries, the scope of the work comprises: the MDVR-KCL dataset as the training corpus; raw audio as the only input representation, with all features extracted by the project’s own code; classical machine learning as the modelling approach; an independent Italian corpus as an external test set; and a file-based prototype with an additional microphone demonstration mode. Live clinical deployment, longitudinal patient tracking, and deep learning approaches are outside the scope of this project.

1.6 Report Organization

The remainder of this report is organized as follows.

Chapter 2 — Background and Literature Review presents the clinical background of PD and its effect on speech production, defines the acoustic measurements used in voice analysis, introduces the classical ML models employed, examines the data-leakage problem that affects much of the published literature, and reviews related work on the datasets used here.

Chapter 3 — Methodology specifies the complete system: the dataset inspection protocol, the preprocessing chain, recording segmentation, the 74-feature extraction stage, the classification pipeline, the subject-independent validation protocol, the pre-declared model-selection rules, the external validation design, and the prototype.

Chapter 4 — Results and Discussion reports the outcomes of every stage: dataset inspection, feature validation, baseline modelling, the model-improvement experiments, the final model, feature importance, the validation-design demonstration, external validation, and the prototype, followed by a general discussion and a dedicated section on limitations and ethical considerations.

Chapter 5 — Conclusion and Future Work summarizes the findings in points and lists recommendations for future development.

Four appendices provide the complete feature inventory, the full experimental results, a reproduction guide for the software repository, and key code excerpts.

Chapter 2

Background and Literature Review

2.1 Overview

This chapter provides the scientific foundation on which the project is built. Section 2.2 summarizes the clinical background of Parkinson’s Disease (PD). Section 2.3 explains how the human voice is produced and how PD alters it. Section 2.4 defines the acoustic measurements used to quantify those alterations, together with the software tools that compute them. Section 2.5 introduces the classical Machine Learning (ML) models employed in this project. Section 2.6 examines the validation problem that affects much of the published literature in this field. Section 2.7 reviews related work and the public datasets used, and Section 2.8 states the research gap that this project addresses.

2.2 Parkinson’s Disease

PD is a chronic, progressive neurodegenerative disorder caused by the loss of dopamine-producing neurons, predominantly in the substantia nigra pars compacta, a region of the midbrain involved in the control of movement. Post-mortem morphometric analysis indicates that this loss is regionally selective and that a substantial proportion of the affected neurons has already degenerated by the time motor symptoms appear [3]. The cardinal motor manifestations are bradykinesia (slowness of movement), muscular rigidity, rest tremor, and, in later stages, postural instability; the disease also produces a wide range of non-motor symptoms [2].

The burden of the disease is substantial and increasing. The Global Burden of Disease

study estimated that about 6.1 million individuals were living with PD worldwide in 2016, compared with 2.5 million in 1990, an increase driven mainly by the growing number of older people [1]. There is at present no cure and no routine laboratory test; the diagnosis is clinical, based on the motor examination and supporting criteria formalized by the Movement Disorder Society (MDS) [2].

Two aspects of this clinical picture motivate the present project. First, because the diagnosis depends on visible motor signs, it is typically made only after the underlying neuronal loss is well advanced [3]. Second, the muscles of respiration, the larynx, and the articulators are affected by the same motor impairment as the limbs, so measurable changes in the voice can accompany — and in some reported cases precede — the visible signs [5, 6]. An inexpensive measurement channel that reflects early motor impairment is therefore of practical interest as a screening-support signal, provided its output is interpreted cautiously and never treated as a diagnosis.

2.3 Speech Production and Hypokinetic Dysarthria

2.3.1 The Source–Filter Model of Voice Production

Human voice production is conventionally described by the source–filter model, illustrated in Figure 2.1. Three stages cooperate:

1. **Power source.** The lungs supply a steady stream of pressurized air through the trachea.
2. **Sound source.** The two vocal folds of the larynx are brought together across the airway. Subglottal pressure forces them apart, and their elasticity, together with aerodynamic forces, snaps them shut again. This cycle repeats quasi-periodically, releasing a train of air puffs. The repetition rate of this cycle is perceived as the pitch of the voice and is called the Fundamental Frequency (F_0); typical adult values lie roughly between 85 and 180 Hz for men and 165 and 255 Hz for women.
3. **Filter.** The pharynx, oral cavity, tongue, teeth, and lips form a resonating tube whose shape changes continuously during speech. This vocal-tract filter emphasizes certain frequency regions and attenuates others, shaping the buzz of the source into distinguishable vowels and consonants.

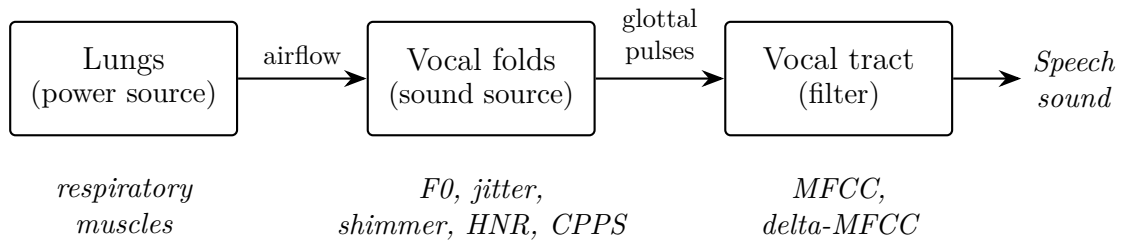


Figure (2.1): The source–filter model of voice production. The lower row indicates which acoustic feature families used in this project characterize each stage.

Two properties of this system matter for the measurements that follow. The rate of vocal-fold vibration determines F0, and because typical F0 ranges differ between adult men and women, absolute pitch values partly encode the speaker’s sex — a potential confounder examined explicitly in Chapter 4. The *regularity* of the vibration determines perceived voice quality: in a healthy voice, successive cycles are nearly identical, while irregular timing or amplitude is heard as roughness, and incomplete closure of the folds is heard as breathiness.

2.3.2 Hypokinetic Dysarthria in Parkinson’s Disease

Dysarthria denotes a group of speech disorders caused by impaired neuromuscular control of the speech apparatus. In their classical perceptual study, Darley, Aronson, and Brown identified distinct dysarthria patterns corresponding to different neurological lesions and associated parkinsonism with *hypokinetic* dysarthria [8]. Its perceptual hallmarks are reduced loudness, monotone pitch (reduced F0 variability), breathy or hoarse voice quality, imprecise consonant articulation, and disturbed speech timing.

These perceptual signs have measurable acoustic correlates, and their prevalence is high. In a sample of 200 PD patients, speech or voice impairment was present in approximately 90% [4]. Quantitative acoustic analysis of 46 early, untreated PD patients found some form of vocal impairment in about 78%, with measures of F0 variability among the most discriminative [5]. Timing is affected as well: compared with healthy speakers reading the same text, PD patients showed altered pause behaviour and a tendency to accelerate their speech rate [9]. A longitudinal case study further suggested that reduced F0 variability can be detected in recordings made about five years before clinical diagnosis [6].

These findings define the measurement targets for this project: pitch variability,

cycle-to-cycle regularity, the balance of periodic and noise energy, articulatory spectral dynamics, and pause behaviour.

2.4 Acoustic Measurement of the Voice

This section defines the acoustic quantities used in the project. All perturbation and cepstral measures are computed with the Praat speech analysis system [10] through its Python interface Parselmouth [11]; spectral features are computed with the librosa library [12].

2.4.1 Fundamental Frequency Statistics

F0 is estimated over short overlapping analysis frames using the autocorrelation method of Boersma [13], which locates the dominant periodicity of the waveform within a configured search range (75–500 Hz in this project, covering adult male and female voices). Frames without detectable periodicity (silences, unvoiced consonants) are excluded. From the resulting F0 track, summary statistics are computed: mean, median, standard deviation, minimum, maximum, range, and the voiced fraction (the proportion of frames in which voicing was detected). Given the monotone character of hypokinetic dysarthria, the dispersion statistics (standard deviation and range) are the most clinically meaningful [5, 6].

2.4.2 Jitter

Jitter quantifies the cycle-to-cycle instability of vocal-fold *timing*. Let T_i denote the duration of the i -th glottal period and N the number of periods. Local jitter is the mean absolute difference between consecutive periods, normalized by the mean period:

$$\text{jitter}_{\text{local}} = \frac{\frac{1}{N-1} \sum_{i=1}^{N-1} |T_i - T_{i+1}|}{\frac{1}{N} \sum_{i=1}^N T_i}. \quad (2.1)$$

Variants used in this project average over small neighbourhoods of periods to reduce sensitivity to isolated measurement errors: absolute local jitter (the numerator of Eq. 2.1,

in seconds), the Relative Average Perturbation (RAP, three-period neighbourhood), and the five-period Perturbation Quotient (PPQ5). In healthy sustained phonation, local jitter is typically below about 1%; impaired neuromuscular control raises it.

2.4.3 Shimmer

Shimmer applies the same idea to cycle *amplitudes* A_i :

$$\text{shimmer}_{\text{local}} = \frac{\frac{1}{N-1} \sum_{i=1}^{N-1} |A_i - A_{i+1}|}{\frac{1}{N} \sum_{i=1}^N A_i}. \quad (2.2)$$

The project uses local shimmer, its decibel form, and the three-, five-, and eleven-period Amplitude Perturbation Quotients (APQ3, APQ5, APQ11). Weak or inconsistent vocal-fold closure produces uneven pulse strength and therefore raised shimmer. Because shimmer is normalized by the mean amplitude, a uniform change of recording gain does not affect it.

2.4.4 Harmonics-to-Noise Ratio

The voice signal is a mixture of a periodic component (the harmonics of F0) and an aperiodic noise component caused by turbulent airflow. The Harmonics-to-Noise Ratio (HNR) expresses their energy ratio in decibels. Using the normalized autocorrelation approach of Boersma, if r_{max} is the height of the normalized autocorrelation peak at the detected period (the fraction of signal power that is periodic), then

$$\text{HNR} = 10 \log_{10} \frac{r_{\text{max}}}{1 - r_{\text{max}}} \text{ dB}. \quad (2.3)$$

A clear voice may exceed 20 dB, while breathy voices score much lower [13]. Incomplete glottal closure — a recognized feature of hypokinetic dysarthria — lowers HNR.

2.4.5 Smoothed Cepstral Peak Prominence

Jitter, shimmer, and HNR presuppose that individual glottal cycles can be identified reliably, which holds for sustained vowels but only approximately for continuous speech.

The Smoothed Cepstral Peak Prominence (CPPS) avoids this requirement. The cepstrum is obtained by taking a second spectral transform of the logarithmic power spectrum; a strongly periodic voice, whose spectrum contains regularly spaced harmonics, produces a sharp cepstral peak at the quefreny of the glottal period. CPPS measures how far this peak rises above the regression line fitted to the smoothed cepstrum, in decibels. A meta-analysis of acoustic correlates of perceived voice quality found cepstral peak prominence measures to be among the strongest and most robust correlates of overall dysphonia, in both sustained vowels and continuous speech [14]. Breathier, noisier voices produce weaker cepstral peaks and therefore lower CPPS.

2.4.6 Pause and Timing Statistics

Speech timing is quantified here with an energy-based segmentation of the recording into speech stretches and silences. Gaps of at least 0.2 s are counted as pauses, from which four features are derived: pauses per minute, mean pause duration, the fraction of total time spent in pauses, and the mean duration of continuous speech stretches. The clinical motivation follows from the altered pause behaviour documented in PD reading tasks [9]. These features have the additional practical advantage of depending on timing rather than on spectral content, which makes them comparatively robust to changes of microphone and recording channel.

2.4.7 Mel-Frequency Cepstral Coefficients and Their Dynamics

The Mel-Frequency Cepstral Coefficients (MFCC) provide a compact description of the short-term spectral envelope — effectively, of the vocal-tract filter configuration. The signal is divided into short frames; for each frame the power spectrum is computed, integrated into overlapping bands spaced on the mel scale,

$$m(f) = 2595 \log_{10} \left(1 + \frac{f}{700} \right), \quad (2.4)$$

which approximates the frequency resolution of human hearing; the band energies are logarithmized; and a discrete cosine transform yields the coefficients, of which the first 13 are retained here, following the representation introduced by Davis and Mermelstein [15]. Delta-MFCC coefficients are the frame-to-frame time derivatives of the MFCCs

and describe how quickly the spectral envelope changes — a proxy for articulatory movement speed. In this project each coefficient trajectory is summarized by its mean and standard deviation over the analysed audio, yielding 26 MFCC and 26 delta-MFCC features. Because MFCCs describe the spectrum in detail, they are also sensitive to the microphone, room, and language — a property with consequences examined in the external validation (Chapter 4).

2.5 Classical Machine Learning Models

Four classifier families, all implemented in the scikit-learn library [16], are used in this project, together with a majority-class baseline that always predicts the more frequent class and therefore represents zero learned knowledge.

Logistic regression models the probability of class membership as a logistic function of a weighted sum of the features. It produces a linear decision boundary, and the learned weights indicate each feature’s direction and strength of influence, making it the most directly interpretable model considered.

The Support Vector Machine (SVM) [17] seeks the decision boundary that separates the classes with the widest possible margin, which favours good generalization from limited data. With the Radial Basis Function (RBF) kernel, the data are implicitly mapped into a higher-dimensional space, allowing the boundary to be curved while the margin principle is retained.

The Random Forest (RF) [18] is an ensemble of decision trees, each trained on a bootstrap sample of the training data with random feature subsets considered at each split; the forest predicts by aggregating the votes of its trees. RFs handle features of mixed scales and nonlinear interactions naturally, resist overfitting through averaging, and provide a ranking of feature importance — a property used in Chapter 4 to interpret the final model.

The Multilayer Perceptron (MLP) is a small feed-forward neural network trained by gradient descent. It is included for completeness as the most flexible of the four families, but flexibility carries risk with only 37 subjects, and its behaviour under a confounder ablation (Chapter 4) illustrates exactly this danger.

The rationale for restricting the project to such classical models rather than deep

learning is threefold: the sample size (37 subjects) is far below what deep networks require; the biomedical setting demands interpretable evidence of *what* the model uses; and the project’s scientific focus is on honest validation, which is clearer to demonstrate with simple, well-understood models.

2.6 Validation Methodology and the Data-Leakage Problem

The value of any reported accuracy depends entirely on how the model was tested. When a dataset contains several recordings — or several excerpts of one recording — from the same person, a random split of those items between training and test sets allows the model to encounter each test speaker during training. The model can then succeed by *recognizing the speaker or the recording*, rather than the disease, and the measured performance no longer predicts behaviour on new people. This phenomenon is a form of data leakage, and in the clinical machine learning literature the distinction is drawn between *record-wise* and *subject-wise* cross-validation: Saeb et al. demonstrated that record-wise validation can dramatically overestimate diagnostic accuracy and argued that the validation scheme must approximate the intended use case — diagnosing *new* individuals [19].

The correct procedure is grouped, subject-independent validation: all data from one person is assigned as a block to either the training or the test side of every split. Where class proportions should also be preserved across folds, stratified grouped K-fold splitting is used. This project enforces subject independence mechanically — a runtime assertion halts execution if any subject ever appears on both sides of a split — and additionally demonstrates the size of the leakage effect empirically on its own data, by evaluating the same pipeline under a deliberately wrong chunk-level split (Chapter 4).

A related, subtler leakage channel concerns preprocessing statistics: if imputation values or feature scaling parameters are estimated on the full dataset before splitting, information about the test data enters training. The methodology of Chapter 3 therefore fits all such operations inside each training fold only.

2.7 Related Work and Datasets

Research on voice-based PD assessment spans two broad recording paradigms. The first uses *sustained vowels*, in which the speaker phonates a steady “ah”; this isolates phonatory function and suits the classical perturbation measures. Little et al. combined dysphonia measures of sustained vowels with an SVM classifier to separate PD speakers from controls [20], and Tsanas et al. extended this line with large feature sets and feature selection, reporting almost 99% classification accuracy on sustained-vowel data — while explicitly noting that their cross-validation did not enforce subject separation and that speaker-specific cues may therefore contribute [21]. The second paradigm uses *connected speech* (read text or spontaneous conversation), which engages articulation and prosody in addition to phonation and is closer to natural speaking conditions [5, 9]; it is the paradigm of the present project.

Two public connected-speech datasets are used in this work. The MDVR-KCL corpus [7] contains smartphone recordings of read text and spontaneous dialogue from 37 English-speaking participants (21 healthy controls and 16 PD patients) and serves as the training corpus; its detailed inspection appears in Chapter 4. The Italian Parkinson’s Voice and Speech database, collected by Dimauro et al. in the context of speech-intelligibility assessment [22], contains recordings of 65 Italian speakers (young healthy, elderly healthy, and PD groups) and serves here exclusively as an external, never-trained-on test set.

Published classification studies on MDVR-KCL and similar corpora frequently report accuracies above 95%. In light of the record-wise leakage mechanism described above [19], and of the demonstration in Chapter 4 that a chunk-level random split inflates this project’s own balanced accuracy from 0.775 to 0.909 without any change to the model, such figures should be interpreted with caution unless subject-independent validation is stated explicitly.

2.8 Research Gap and Project Contribution

From the literature reviewed above, three gaps emerge. First, many reported results rest on validation schemes that do not separate speakers, leaving their practical meaning uncertain [19, 21]. Second, cross-dataset evaluation — testing a trained system on an entirely different corpus, language, and recording channel — is rare, although it is the

most direct measure of the generalization that a screening application would require. Third, reported systems seldom address how their output should be presented given the uncertainty involved; a bare class label overstates what a small-corpus model knows.

This project addresses all three gaps within an undergraduate scope: it enforces subject-independent validation mechanically and quantifies the leakage effect on its own data; it performs an unadapted external validation on a second corpus in a different language; and it delivers a prototype whose output includes an explicit inconclusive band, recording-condition warnings, and mandatory non-diagnostic wording. The complete methodology follows in Chapter 3.

Chapter 3

Methodology

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Chapter 4

Results and Discussion

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Chapter 5

Conclusion and Future Work

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Appendix A

Complete Feature Inventory

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Appendix B

Full Experimental Results

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Appendix C

Software Repository and Reproduction Guide

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Appendix D

Key Code Excerpts

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